



Lab Report: Young's Double Slit Interference

Student Name: _____

Date: _____

1. Theory

Constructive interference (bright fringes) occurs at:

$$d \sin \theta = m\lambda$$

Fringe separation (Δy) is related to screen distance (L): $\Delta y = \frac{\lambda L}{d}$.

2. Data Collection

Trial	Slit Separation (d)	Wavelength (λ)	Fringe Space (Δy)	Observations
1		450 nm (Blue)		
2		650 nm (Red)		
3		650 nm (Red)		

3. Analysis Questions

1. Comparing Trial 1 and 2, which color produces a wider fringe separation? Why?
2. If you increase the distance d between the slits, what happens to the interference pattern?